case, the swirl is so significant that, mathematical analysis shows, not only space but also time would get caught up in the whirlpool. Roughly speaking, the spinning twists the time direction on its side, so that circular motion around the cylinder takes you to the past. If your rocket ship encircles the cylinder, you can return to your starting point in space *before* you embark on your journey. Certainly, no one can build an infinitely long spinning cylinder, but this work was an early hint that general relativity might not prohibit time travel to the past.

Gödel's paper also investigated a situation involving rotational motion. But rather than focusing on an object rotating within space, Gödel studied what happens if all of space undergoes rotational motion. Mach would have thought this meaningless. If the whole universe is rotating, then there's nothing with respect to which the purported rotation is happening. Mach would conclude, a rotating universe and a stationary universe are one and the same. But this is another example in which general relativity fails to fully conform to Mach's relational conception of space. According to general relativity, it does make sense to speak of the entire universe's rotating, and with this possibility come simple observational consequences. For example, if you fire a laser beam in a rotating universe, general relativity shows that it will appear to travel along a spiral path rather than a straight line (somewhat like the path you'd see a slow-moving bullet follow if you fired a toy gun upward while riding a merry-go-round). The surprising feature of Gödel's analysis was his realization that if your rocket ship were to follow appropriate trajectories in a spinning universe, you could also return to your place of origin in space *before* the time of your departure. A rotating universe would thus itself be a time machine.

Einstein congratulated Gödel on his discovery, but suggested that further investigation might show that solutions to the equations of general relativity permitting travel to the past run afoul of other essential physical requirements, making them no more than mathematical curiosities. As far as Gödel's solution goes, increasingly precise observations have minimized the direct relevance of his work by establishing that our universe is not rotating. But van Stockum and Gödel had let the genie out of the bottle; within a couple of decades, yet more solutions to Einstein's equations permitting time travel to the past were found.